

System Administration Lab

Open Ended Experiment
(To be included in Continuous Internal Evaluation)

Group Number	7
Project Title	File Test Shell Script
Platform	Linux Terminal
Experiment Type	Open Ended Experiment

1. Aim

To write, execute, and demonstrate a shell script that accepts a file name from standard input and performs file tests for existence, readability, and writability.

2. Objective

The objective is to understand the use of file-test operators in a shell script and execute the required file tests correctly in the Linux terminal.

3. File Tests

Sl. No.	Operation	Shell Test
1	Check file existence	[-e "\$filename"]
2	Check whether file is readable	[-r "\$filename"]
3	Check whether file is writable	[-w "\$filename"]

4. Algorithm

- Start the Linux terminal.
- Create/open the shell script.
- Accept the file name from standard input.
- Check whether the file exists.
- If the file exists, check whether it is readable.
- Check whether the file is writable.
- Display the corresponding result for each test.
- Stop the experiment.

5. Shell Script

```
#!/bin/bash

echo "Enter the file name:"
read filename

if [ -e "$filename" ]; then
    echo "File exists."

    if [ -r "$filename" ]; then
        echo "File is readable."
    else
        echo "File is not readable."
    fi
fi
```

```
if [ -w "$filename" ]; then
    echo "File is writable."
else
    echo "File is not writable."
fi
else
    echo "File does not exist."
fi
```

6. Execution

Open a Linux terminal and save the program as **filetest.sh**. Give execute permission and run the script.

```
chmod +x filetest.sh
./filetest.sh
```

7. Sample Execution

```
$ ./filetest.sh
Enter the file name:
sample.txt
File exists.
File is readable.
File is writable.
```

Sample execution when the file does not exist:

```
$ ./filetest.sh
Enter the file name:
abc.txt
File does not exist.
```

8. Expected Output

The terminal should display whether the entered file exists. If the file exists, the script should also display whether the file is readable and writable.

9. Result / Observation

The file test shell script was written and executed successfully in the Linux terminal. The experiment demonstrated file existence, readability, and writability tests using shell scripting.

10. Guidelines for the Open Ended Experiment

- The experiment is performed by Group 7.
- The script should be executed and tested in the Linux terminal.
- A suitable file name should be entered during execution.
- Students should understand the purpose and syntax of the file-test operators used.
- The final observation should include the execution results.

11. Evaluation Criteria

Sl. No.	Criteria
1	Understanding of shell script and file tests
2	Correct execution of the script
3	Understanding of file-test syntax and output
4	Presentation of results
5	Team participation

Group Number: 7

Instructor's Signature: _____

Date: _____